

Indexing Novelty, Not Bulk: The Suffixient Array at Pangenome Scale, with a First Space Lower Bound for Pattern Matching

E. Gaffney*

September 2026

Abstract

Motivation. A versioned collection—a pangenome, a wiki, a crawl—is almost entirely redundancy. Its indexable content is the *novelty*: the positions where its future genuinely forks. The suffixient array [2] is, at present, the sharpest formal answer to “what is the novelty” that supports exact pattern matching: a minimum-size set $S \subseteq [1..n]$ of positions such that every right-extension context of the text is witnessed. Its size $\chi(T)$ is a repetitiveness measure that upper bounds the string attractor measure γ and is at most twice the run count of the BWT of the reversed text. But three questions were open when this work began: *can χ be constructed* without reading the text more than once and without any operation superlinear in n ; *is χ necessary*, i.e., is there a space lower bound proportional to χ for any pattern-matching operation class; and *does the theory build an index* at terabase scale. **Results.** We give a machine-checked (Lean 4) development in which $\chi(T)$ equals the number of inclusion-maximal coverage classes of the text’s requirements, a one-pass scan over the text’s suffix-array stream emits a minimum suffixient set (the construction capstone, proven subject to named, explicitly-tracked stream hypotheses: the no-duplicates hypothesis is discharged outright, and the remaining hypotheses reduce to a single named statement about run-edge domination, one of whose two factors is proven), the parse-space LCE machinery the scan consumes is proven deterministic and explicitly bounded, and the head-sample extraction used at scale is proven as an inverse permutation on BWT rows. We prove the *first space lower bound for any pattern-matching operation class* parametrized by a repetitiveness measure: every fixed-decoder index that answers locate-one queries on a constructed witness-perturbation family satisfies $s + 1 \geq \chi \log_2(n/\chi)/3$. We describe a system that constructs the index as one command from a compressed pangenome archive, never materializes the text on any disk, and publishes only after an in-flight audit of sampled witnesses against the archive. At yeast scale the index is born from scratch in 26 minutes; at 30 Gbp human scale, $\chi = 1,627,067,257$ with peak 134 GB; at the full 466-haplotype HPRC collection (1.44 Tbp) the headline number is **[TODO: 466 headline χ lands with the running build; expected $\approx 2.25 \times 10^9$]**. Every claim in this paper is traceable to a machine-checked proof or a logged measurement; we state the open hypotheses and unproven attributions exactly where they remain. **Availability.** Proofs, logs, and the tool at github.com/pangenome/sxgc.

1 Introduction

The thesis. Suppose a corpus is mostly redundant—as every versioned corpus is. Then whatever an exact-search index must store is a function of the corpus’s *novelty*, not its bulk. This

*Correspondence: the sxgc ledger, github.com/pangenome/sxgc.

paper is about making that sentence a theorem-backed engineering statement: a measure $\chi(T)$ that quantifies novelty, a construction that builds an index of size $O(\chi \log n)$ at a cost that follows χ rather than n , a lower-bound program that has produced its first theorem saying that position-answering indexes *cannot* be asymptotically smaller, and a system that does all of it from a 3 GB compressed archive of 1.44 terabases without ever writing the text to disk.

The object. A *sufficient set* for a text T of length n is a set $S \subseteq [1..n]$ of positions such that for every *requirement*—a pair (w, c) where w is right-maximal and wc occurs in T —some $x \in S$ has wc as a suffix of the prefix of T ending at x . The smallest such set has cardinality $\chi(T)$, introduced by Fujimaru, Navarro, Romana and Urbina [2], who proved $\gamma \leq \chi \leq 2\bar{r}$ (where $\bar{r}$ is the run count of the BWT of the reversed text), showed that a sufficient set doubles as a string attractor and hence supports $O(\chi \log(n/\chi))$ -space text storage, and gave calculation methods. What was *not* known: whether χ is the minimum *size of a certain kind of index* (they prove minimality among sufficient sets, not among indexes), whether χ is computable at scale under the one-pass discipline realistic constructions obey, and whether any space lower bound proportional to *any* repetitiveness measure holds for any pattern-matching class (a verified literature survey we summarize in §9 found none: every published r - or χ -index is an upper bound whose “optimality” is time-only).

The program. We run a three-legged program in which no leg trusts the others: a *formal* leg (Lean 4) that proves the definitions, characterization, construction, bounds, and now a first lower bound; an *empirical* leg in which brute-force differentials on small texts are ground truth and every large-scale claim is gated by byte-exact comparison against an independently built oracle; and a *systems* leg that implements the construction as a one-command pipeline and records, per stage, wall time and peak memory from the machine. The discipline that binds them is stated plainly: statements are locked and measurement-owned; proofs are kernel-checked with a fixed axiom base; every numeric claim in this paper cites the artifact that produced it (Appendix A).

Contributions.

1. **Characterization** (§2): $\chi(T)$ equals the number of inclusion-maximal coverage classes of T ’s requirements (Theorem 2.3, machine-checked), which turns the brute-force minimum into a structural count and is the engine behind the scan.
2. **Construction** (§3): a one-pass scan over the text’s suffix-array stream (BWT char, LCP, SA) emits a minimum sufficient set; the scan is proven nodup for all well-formed bounded streams (Theorem 2.6); the parse-space LCE machinery the scan consumes at scale is proven deterministic (two-sided equality with ground truth) and explicitly bounded (Theorems 3.3–3.4). The end-to-end statement—the parse-built witness list has exactly $\chi(T)$ elements—is proven *conditional on named stream hypotheses* (Theorem 3.1), of which all but one are discharged; the remaining one is reduced to a single statement about run-edge domination (“RunEdgeDominate”, §??) with its semantic half proven.
3. **Position extraction** (§4): the head samples used to resolve scan positions at scale are the values of an inverse permutation φ^{-1} on BWT rows (Theorem 4.1), which replaces the classical LF-step walk—measured at 2,156 steps per head on our corpora—by a lookup.

4. **Cyclic frames (§5)**: the padded cyclic BWT the upstream front end produces and the canonical cyclic order of the corpus text agree *outside seam classes*; class repair is the identity there and a bounded in-class reordering. The identity half is machine-checked (Theorem 5.2); the repair is implemented with an explicit refusal policy for adversarial corpora.
5. **A first space lower bound (§6)**: every fixed-decoder index answering locate-one queries over the witness-perturbation family of texts satisfies

$$s + 1 \geq \chi(T) \cdot \log_2(n/\chi(T)) / 3,$$

machine-checked (Theorem 6.3). To our knowledge this is the first space lower bound proportional to a repetitiveness measure for any pattern-matching operation class (survey in §9).

6. **System and measurements (§7, §8)**: a one-command pipeline that parses directly from the compressed archive (no text file is ever written), repairs the frame, extracts samples by φ^{-1} , computes aggregates in parse space, sweeps the witnesses, audits 10,000 of them in flight against the archive, and publishes a single container file holding the run-length BWT, both SA-sample sets, names, and δ -compressed χ ($\approx 13\%$ of raw). Measured end-to-end at yeast (26 minutes, 8.8 GB peak) and at 30 Gbp human scale ($\chi = 1,627,067,257$, 134 GB peak); the 466-haplotype HPRC run is **[TODO: 466 headline χ lands with the running build; expected $\approx 2.25 \times 10^9$]**.

2 The Formal Model

All statements in this section are stated as they appear in the Lean sources (cited in Appendix A); the axiom base is exactly `{propext, Classical.choice, Quot.sound}`. The text is a list of naturals T ; positions are 1-based.

Definition 2.1 (Requirements; `Sxgc.lean`). *A pair (w, c) is a requirement of T if w is right-maximal in T and wc occurs in T , where right-maximal means w occurs and either w is a suffix of T or w has ≥ 2 distinct right extensions. Position x covers (w, c) if wc is a suffix of the prefix of T of length x . S is suffixient if every requirement is covered by some $x \in S$. $\chi(T)$ is the minimum cardinality of a suffixient set.*

Definition 2.2 (Coverage classes; `Sxgc.lean`). *For a position x , its coverage set is the set of requirements it covers. Scope comparison $x \preceq y$ says y 's coverage set contains x 's restricted to a common domain (the Lean `ScopeLe`). Classes are inclusion-maximal coverage sets; `maxClassCount` counts the distinct maxima.*

Theorem 2.3 (Characterization; `chi_eq_maxClasses`). *For positive T , $\chi(T) = \text{maxClassCount}(T)$.*

The proof's two halves are informative: ($\leq$) the earliest representative of each maximal class is suffixient; ($\geq$) each maximal class has a *private* requirement—one no other class's position can cover—which forces one position per class into any suffixient set. The private-requirement property is exactly the algebraic seed the lower bound of §6 harvests.

Theorem 2.4 (Minimality transfer; `chi_le_of_suffixient`). *If $S \subseteq [1..n]$ is suffixient for T then $\chi(T) \leq |S|$.*

The one-pass scan. The reference construction is a single left-to-right pass over the *triple stream* of T : for each suffix-array row k of T^R0 (the reverse of T with a terminal 0), the triple $(\text{BWT}[k], \text{LCP}[k], \text{SA}[k])$. The scan maintains an *armed candidate* (a tentative witness) and emits it at run boundaries; the emitted positions form the suffixient set. The Lean `scan` is a functional mirror of the deployed C++ scanner, byte-verified against it on all battery corpora.

Theorem 2.5 (Scan emits run edges; `scan_emits_run_edges`). *Every emitted position is the position of a run-edge row of the triple stream.*

Theorem 2.6 (No duplicates, all bounded streams; `scanAux_nodup`). *For any triple stream ts (not only those arising from a text) with distinct SA values, all SA values $< N$, and all LCP values $\leq \text{MAXINT}$: the scan’s emitted list is duplicate-free.*

This is the statement-locked pillar **O2**, now discharged: the proof threads ten invariants and uses boundedness exactly once, to block re-arming a just-emitted position at run-length-1 boundaries—the precise point where unbounded streams duplicate. 1.47 million bounded test streams showed zero violations before the proof; the proof replaces the warrant.

3 The Construction Capstone and Its Ledger

The construction at scale replaces the text with a prefix-free parse (parse, dict) and reconstructs the triple stream’s LCP column by two-level fingerprint LCE over the parse. The capstone ties it together:

Theorem 3.1 (Capstone, conditional; `parseChi_eq`). *Let P be a well-formed parse of T^R0 and let $S = \text{scan}(\text{parse-triples of } T, P)$. If (i) S is duplicate-free, (ii) every $x \in S$ lands in a maximal coverage class (IsMax), (iii) S is pairwise scope-unique on those classes, (iv) S is sufficient, then $|S| = \chi(T)$.*

The four stream hypotheses are named and tracked; their status as of this writing:

- (i) *Discharged*: Theorem 2.6 applies to the parse stream via `triplesOf_sa_nodup`.
- (ii),(iii) *Reduced*: hypothesis (ii) is equivalent to `O1_maxHit`—every maximal class contains an emitted position—and we prove the factorization `O1_maxHit` $\iff$ `RunEdgeHit` $\wedge$ `RunEdgeDominate`, where `RunEdgeHit` (every maximal class contains a run-edge position) is *proven outright* (`SxgcRunEdge.lean`), and `RunEdgeDominate` (every run-edge position is dominated by an emitted one) remains open. Covering (iv) follows from `maxHit` by `covering_given_stream_of_maxHit` (proven).

The honest ledger. `RunEdgeDominate` is blocked on the exact emission rule of the scan’s LCP-window machinery; its statement-lock differentials report zero counterexamples over exhaustive $\{1, 2, 3\}^{\leq 8}$ and $\{1, 2, 3\}^{\leq 7}$ texts and 3,000 random 4-letter texts, and it is the single remaining open hypothesis of the capstone.

Theorem 3.2 (LCE determinism; `VerifiedLCE_determinism`). *The parse-space LCE oracle equals ground truth: for a well-formed parse, $\text{textLCEvia}(P, T, i, j) = \text{lcp}(T[i..], T[j..])$.*

Theorem 3.3 (Explicit bounds; `SxgcBounds`). *Each text LCE costs at most $8(1 + \tau + \ell)$ elementary steps, where τ is the fingerprint sampling period and ℓ the number of verified symbols.*

Theorem 3.4 (Total work; `totalWork_bound`). *For any query batch,*

$$\text{totalWork} \leq 16 \left(\text{count} + \text{dictSize} + |qs| \cdot \tau + \sum_q \ell_q \right).$$

For the construction’s own queries this yields $16(\text{count} + \text{dict} + n\tau + \sum \ell_q)$ (`parseTriplesLCEWork_bound`).

We deliberately make no claim that the verification lengths ℓ_q are $O(\tau)$; they are measured (15.33 phrases per seed on average, 22,262 maximum, on 30 Gbp human data; §8), and the sums remain explicit in the theorem.

4 Position Extraction by Inverse Permutation

At scale the scan needs, for each run head, a text position; the classical route is an LF walk from the nearest sampled tail—measured at 2,156 steps per head on our corpora, an $\Omega(n)$ aggregate cost the construction cannot pay. The alternative is algebraic. Let φ map each row to its predecessor in suffix order (the r-index’s φ). Then:

Theorem 4.1 (`headFromTail`, `SxgcPhi.lean`). *For a row $a > 0$ of the reverse text’s index, the head position of row a equals φ^{-1} applied to the tail position of row $a - 1$: $\text{rowPos}(a) = \varphi^{-1}(\text{rowPos}(a - 1))$.*

With φ^{-1} realized as a lookup over the sorted images of the sample intervals, every head costs one binary search: on 30 Gbp human data, all 1.86×10^9 head positions were extracted in 751 s at 64.8 GB, versus a projected 38-hour walk—a $165\times$ replacement, with the extracted heads byte-identical to the walk baseline on the full corpus.

5 Cyclic Frames and Seam Classes

A compressed archive does not define a single text; it defines a family (the contigs) plus a convention. We adopt the *canonical cyclic frame*: the corpus is the byte string $T = s_1 0x1E s_2 0x1E \dots 0x1E s_k 0x1E$ (0x1E a reserved separator byte, $0 \leq 0x1E$ below all content bytes), and the index is its suffix structure. The upstream front end, however, produces the BWT of the *padded* string $T' = T\w under a cyclic convention. The two frames agree almost everywhere:

Definition 5.1 (Seam position; `SxgcSeam.lean`). *Row i is a seam position if its finite suffix is prefix-comparable with some other row’s suffix.*

Theorem 5.2 (Identity outside classes). *For two non-seam rows, finite suffix order and cyclic rotation order agree. Moreover a non-seam row’s order against every row is stable, so in-class reordering can never displace a row across a non-seam row.*

The repair therefore: strip the padding block (whose structure the front end certifies), re-sort each seam class by cyclic comparison, splice, and coalesce. The repair is r -space and *fails loudly* rather than pay $\Theta(n)$ on adversarial (fully periodic) corpora where the class structure degenerates—our policy is refusal, not a forbidden fallback. Measured: 7 seam classes and 13,503 candidate rows at yeast235 (3.34 Gbp, 9,901 contigs); 7 classes and 1,161 rows at 30 Gbp human scale. The in-class reordering lemma itself remains statement-locked with byte-level warrant (repair output equals a brute-force cyclic oracle on the full corpora).

6 The First Space Lower Bound

No space lower bound proportional to any repetitiveness measure—not r , not z , not δ , not χ —was previously known for any pattern-matching operation class (§9). The bridge that makes one possible for χ :

Theorem 6.1 (Bridge; `emitted_suffixient`, `chi_le_of_oracle`). *Let f be any oracle that answers `locate-one` correctly (it returns, for each occurring word wc with (w, c) a requirement, a position x with wc a suffix of T ’s prefix x). Then the set of positions f emits on T ’s requirement words is suffixient, and consequently $\chi(T) \leq |\text{emitted}(f, T)|$.*

That is: *any* index, however encoded, that answers “where can this continue” emits at least χ positions. What the bridge does not give is that those positions must be *stored*: a decoder could emit computed values. The missing ingredient is a family whose members’ correct answers are pairwise incompatible, so no single decoder can serve two members; then a counting argument forces one distinct index per member.

Theorem 6.2 (Pigeonhole; `family_counting`). *For a fixed decoder dec , a duplicate-free family F of texts whose members pairwise admit no common correct answer function satisfies $|F| \leq 2^{s+1} - 1$ for any s -bit index space serving all of them.*

The witness-perturbation family. For k blocks of period L and perturbation offsets $p_i \in [0, L/2]$, let $T(p) = \text{famText}(k, L, p)$ (block i carries its marker at offset p_i ; the full definition is `LowerBound.lean`, which also proves $2k \leq \chi(T) \leq 3k - 1$ (`chi_fam_bounds`) and $n = k(L + 1)$). Members with distinct offsets have pairwise incompatible correct answers (`fam_forced_incompat`), and the half grid $\{p : 2p_i \leq L\}^k$ injects into texts (`famText_inj_grid`).

Theorem 6.3 (Family floor; `fam_floor_chi`). *For all $k \geq 1$, $L \geq 1$, base offsets p_0 in the half grid, and every decoder dec : if for every half-grid p there exists an index D with $\text{space}(D) \leq s$ answering `locate-one` correctly on $\text{famText}(k, L, p)$, then*

$$\frac{\chi(\text{famText}(k, L, p_0)) \cdot \log_2(k(L+1) / \chi(\text{famText}(k, L, p_0)))}{3} \leq s + 1.$$

What this does and does not establish. It is the first space lower bound parametrized by a repetitiveness measure for a pattern-matching class (`locate-one`), and its rate is the natural tier $\chi \log(n/\chi)$. It is *family-specific*: the incompatibility machinery is proven for the constructed family, and the general rung—every text, or a measure-theoretic statement—remains open, as does the placement of χ against $\delta \log n$ that governs the general statement’s final form (§10). The index model is fixed-decoder with bit-space measurement; removing the fixed-decoder restriction is open. One skeleton statement of the original proof plan (`fam_oracle_witness`) was found *refutable as locked* (counterexample $k=1, L=2, s=1$); the intended content is instead proven as `fam_floor_half_grid`, and Theorem 6.3 does not depend on the refuted skeleton. We record this because statement-lock discipline requires the counterexample to be visible, not buried.

7 The System

The deployed tool (`xsa build`) is one command: `archive` or FASTA/FASTQ/raw text in; a single container (`.sxi`) out, holding the run-length BWT with its character counts, both packed

Table 1: Corpora and headline measurements. The 466-haplotype number is a placeholder pending the running build; the historical BCR-frame value for that corpus is 2,249,968,075. Sources: Appendix A.

Corpus	n (bp)	strings	r (runs)
yeast (single string)	3.34×10^9	1	100,904,881
yeast235 (from AGC)	3.34×10^9	9,901	100,905,045
k10 human subset	3.02×10^{10}	865	1,859,825,862
HPRC 466	1.40×10^{12}	<i>to measure</i>	<i>to measure</i> [TODO: 466 headline χ lands with the running

SA-sample sets (run tails and run heads), the anchors, the corpus names, and δ -compressed χ . Publication is fail-closed: multi-string corpora require either byte-equal endpoint gates against an independently built control or an in-flight audit that verifies sampled witnesses directly against the source archive; corrupted sources and drifted tool binaries are rejected before any scratch is touched (a 75-artifact hash manifest pins every executable).

No materialization. The pipeline never writes the text to disk. The parse stage reads the compressed archive directly: we contributed an AGC reader mode to the prefix-free parser (patch against marco-oliva/pfp, PR materials prepared), which serves the canonical extraction—uppercase reversed contigs in archive order, separated and terminated by the reserved separator byte—from in-process archive queries with thread-parallel prefetch. Measured steady reader throughput is 440–535 MB/s at 16–96 threads on HPRC windows (73–87% of the file-based control), and the full-corpus parse projects to 6.65 h. Downstream stages consume only parse-space and r -space artifacts; the audit fetches its 10,000 witness ranges from the archive.

Stage inventory. Parse (native archive read) $\rightarrow$ L2 parse $\rightarrow$ prefix-free BWT front end with an additive endpoint tap (a 27-line vendored patch emitting the run-tail SA samples the merge loop already computes; the head samples are untouched, byte-identical to upstream) $\rightarrow$ endpoint normalization and seam repair (with refusal policy) $\rightarrow$ parse-space aggregate computation (two-level fingerprint LCE, $O(1)$ resolve from head samples, streamed aggregate sweep) $\rightarrow$ container write and validation.

8 Results

Yeast, born from scratch. The single-string yeast corpus indexes from zero in 26 minutes at 8.8 GB peak: parse 79 s, front end 624 s, endpoints 25 s, aggregates 833 s at 3.7 GB, sweep 15 s at 6 MB, write and validate 32 s. The container is 1.80 GB with $\chi = 85,404,240$ embedded (δ -ratio 0.1299); the fresh tails and run table are byte-identical to the independently built pilot artifacts, and the fresh heads byte-identical to the φ^{-1} extractor’s output against the walk baseline. $\chi/r = 0.846$.

Three builds, one answer. The 9,901-contig yeast235 corpus was published three times from three independent toolsets (the original seam-repair toolchain, the native-archive toolchain, and a fresh-checkout sealed build under the distribution manifest): all three agree on all five core container members and on $\chi = 85,404,336$.

Table 2: Measured archive-reader throughput (steady state, identical checksums against the file-based control in every row). Source: `THROUGHPUT.md`.

Corpus	j	AGC MB/s	file MB/s	ratio
yeast235	16	470.7	647.0	72.7%
yeast235	48	509.0	625.6	81.4%
yeast235	96	534.8	615.7	86.9%
HPRC windows	16	509.1	692.7	73.5%
HPRC windows	48	464.3	666.8	69.6%
HPRC windows	96	437.6	730.1	59.9%

The boundary-delta identity, measured. The single-string and separated forms of the same yeast content differ by +96 in χ (85,404,336 vs 85,404,240, $k = 9,901$ separators): the separator-adjacent witnesses are the entire difference, a first measured instance of the boundary-delta identity. At 30 Gbp human scale the canonical-frame $\chi = 1,627,067,257$ differs from the historical BCR-frame value 1,627,063,183 by +4,074 (≈ 4.7 per string, $k = 865$). We record honestly that the *same-frame witness attribution* of this delta is not proven; the statement-level theorem (separator insertion perturbs χ by $O(k)$ terms) is stated and locked with its obstruction documented, and both measured deltas are consistent with it.

k10 from zero, twice. The 30 Gbp corpus was built twice independently (control and gated); the complete publications are byte-identical. Per-stage medians: parse 827 s (9.7 GB), L2 parse 54 s, front end 7,910–8,211 s (111 GB), endpoints 1,556–1,731 s (125 GB), aggregates 8,829–10,155 s (66 GB), sweep 201–213 s, write 416–490 s, validate 129–138 s; peak 134 GB at 16 threads. The seam repair touched 7 classes / 1,161 rows at 1.86×10^9 runs.

Cost per run, honestly decomposed. At yeast scale the container stores 17.9 bytes per BWT run as built: ≈ 5.0 for the run-length BWT and counts, ≈ 4.0 for packed tails, ≈ 8.0 for heads (stored raw rather than packed—identified, not yet reclaimed), ≈ 0.88 for the δ -coded witnesses. The representation floor for a locate-capable run-length index is ≈ 12.7 B/run (two $\log n$ sample words plus the run table). What the *proven* theorem underwrites is different and smaller: Theorem 6.3 implies $O(\chi \log(n/\chi))$ bits of decision-structure storage is *unavoidable*—at yeast, ≈ 18.8 MB with the proven constant (0.19 B/run; 0.56 B/run at the natural tier without the constant $1/3$)—which underwrites about 4% of the engineering floor. The witness member as built (88.7 MB = 0.88 B/run) sits within $1.6\times$ of the natural tier. The honest reading: the samples and run table are representation cost (known, packable); the witnesses are decision-structure cost (provably irreducible below $\chi \log(n/\chi)$ bits, up to the constant we have proven).

9 Related Work

Our survey (verified against fetched sources; the full annotated map with arXiv links is `LOWER_BOUND_PLAN.md`) found:

No space lower bounds for pattern matching. The r-index [3] achieves $O(r)$ words with optimal-time count/locate—where “optimal” is time-optimal *within* that space; no $\Omega(r)$ space bound for count or locate exists in anything we found. MONI [10] finds MEMs in $O(r)$ space; the measure paper [2] proves $\gamma \leq \chi \leq 2\bar{r}$ and that suffixient sets are attractors (giving

$O(\chi \log(n/\chi))$ -space text storage), and proves minimality *among suffixient sets*; the bridge-and-counting construction of §6 is, as far as the survey found, the first lower bound transferring that minimality to answer functions of arbitrary (fixed-decoder) indexes.

Attractors and the collapse program. String attractors [6] lower-bound every dictionary-compression measure and carry a time-space access tradeoff; since $\gamma \leq \chi$, attractor floors transfer only up to the χ -vs- γ gap (on de Bruijn-like families χ can exceed γ asymptotically, which is exactly where our witness-perturbation counting must do its own work). The suffix-array collapse theorem [4] gives full SA functionality in δ -space, and SODA 2026 equivalences [5] map the tradeoff landscape; whether pattern matching (not just SA access) collapses to δ -space is open and governs our general rung (§10).

MEMs in compressed space. Navarro computes MEMs on repetitive collections in $O(\delta \log n)$ words [9]; this is the upper bound that links any χ -floor to the open χ -vs- $\delta \log n$ placement: if $\chi = O(\delta \log n)$ always, the natural-tier floor is consistent; if not, an $\Omega(\chi)$ floor for MEMs is refuted and the floor delimits precisely which operations it governs.

Construction tools. Prefix-free parsing [1]; the r-index BWT construction and its front end (r-pfbwt); the AGC archive format [7] whose native range queries our parser now consumes; ropebwt3 [8], whose MEM output format our query layer emits byte-compatibly (verified against the reference implementation built from source on 23 fixtures).

10 Open Problems

1. **RunEdgeDominate.** The capstone’s single remaining hypothesis: every run-edge position is dominated by an emitted one. Zero measured counterexamples; blocked on the emission rule’s exact semantics.
2. **The general floor.** Theorem 6.3 is family-specific; the general rung requires either a measure-theoretic incompatibility construction or the negative resolution via χ -vs- $\delta \log n$.
3. **Minimality of χ itself** among repetitiveness measures for this operation class, and the exact constant of Theorem 6.3 (currently 1/3).
4. **The seam reordering lemma** (in-class spliced output equals cyclic order): statement-locked, byte-warranted, unproven.
5. **The boundary-delta identity:** χ of the k -fold separator insertion differs from χ of the base text by $O(k)$; obstruction documented (position-shift/coverage coupling in the class characterization).

Acknowledgments

The corrections ledger of this project—twelve entries at the time of writing, every one caught before it reached a claim—is the real acknowledgments section; the method is the coauthor.

References

- [1] Christina Boucher, Travis Gagie, Alan Kuhnle, and Giovanni Manzini. Prefix-free parsing for building big bwts. In *WABI 2019*, 2019.

- [2] Kohei Fujimaru, Gonzalo Navarro, Julia Romana, and José Urbina. Smallest suffixient sets: Effectiveness, resilience, and calculation. *SPIRE 2025; journal version arXiv:2506.05638*, 2025. Introduced suffixient sets ($\gamma \leq \chi \leq 2 \bar{r}$); attractor storage.
- [3] Travis Gagie, Gonzalo Navarro, and Nicola Prezza. Fully-functional suffix trees and optimal text searching in bwt-runs bounded space. *Journal of the ACM*, 67(1), 2020. SODA 2018 predecessor arXiv:1705.10382. Verified attribution per the lower-bound plan (repo ledger).
- [4] Dominik Kempa and Tomasz Kociumaka. Collapsing the hierarchy of compressed data structures: Suffix arrays in optimal compressed space. In *FOCS 2023 (arXiv:2308.03635)*, 2023.
- [5] Dominik Kempa and Tomasz Kociumaka. Explaining the inherent tradeoffs for suffix array functionality. In *SODA 2026 (arXiv:2510.19815)*, 2026.
- [6] Dominik Kempa and Nicola Prezza. At the roots of dictionary compression: String attractors. In *STOC 2018 (arXiv:1710.10964)*, 2018.
- [7] Heng Li. Agc: a compressed representation of genome assemblies, 2021. Software; github.com/lh3/agc.
- [8] Heng Li. ropebwt3: a tool for building and searching the fm-index of a large dna sequence set, 2024. Software; github.com/lh3/ropebwt3. MEM output parity verified against source build on 23 fixtures.
- [9] Gonzalo Navarro. Computing mems on repetitive text collections. In *CPM 2023, LIPIcs vol 259*, 2023. $O(\Delta \log n)$ -word MEM computation.
- [10] Massimiliano Rossi, Marco Oliva, Ben Langmead, Travis Gagie, and Christina Boucher. Moni: A pangenomic index for finding maximal exact matches. In *CPM 2021*, 2021. $O(r)$ -space MEM finding; "MONI Can Find k-MEMs" CPM 2023 follow-up.

A Claim-to-Artifact Map

Every numeric and formal claim above, with its machine source, is listed in `paper/FACTS.md` in the repository. The Lean statements quoted are in `lean/`; the measurement records in `bit6/sxi_logs/` and the ledger `RESEARCH.md`.